AS MATHEMATICS - PURE 1 - DAY 13

Integration

Antiderivatives, constants and areas

YOUR AIM TODAY

Reverse differentiation and evaluate definite integrals.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

LEARN FROM ZERO

The ideas you need

Integration rebuilds a function from its rate of change.

- Integral $x^n\,dx = x^{(n+1)}/(n+1) + C$ for n not equal to -1.
- Add C for an indefinite integral.
- Use a given point to find C.
- For a definite integral, evaluate upper minus lower.
- Area may require splitting where the curve crosses the axis.

MEMORY RULE

Increase the power by one, divide by the new power.

Indefinite

WORKED STEPS

Integral $(6x^2-4)\,dx = 2x^3-4x+C$.

Definite

WORKED STEPS

Integral from 0 to 2 of $3x^2\,dx = [x^3]_0^2 = 8$.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Integrate 4x^3.	
2. Integrate 6x^2-2x+5.	
3. Given dy/dx=2x+3 and y=5 at x=1, find y.	
4. Evaluate integral 0 to 3 of 2x dx.	
5. Evaluate integral 1 to 2 of (3x^2+1) dx.	
6. Find area under y=4-x^2 from x=0 to x=2.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	x^4+C.
2	2x^3-x^2+5x+C.
3	y=x^2+3x+C; 5=1+3+C, so C=1. y=x^2+3x+1.
4	[x^2]0 to 3=9.
5	[x^3+x]1 to 2=(10)-(2)=8.
6	[4x-x^3/3]0 to 2=8-8/3=16/3.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.